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Design Strategies for Mycelium-Based Composites

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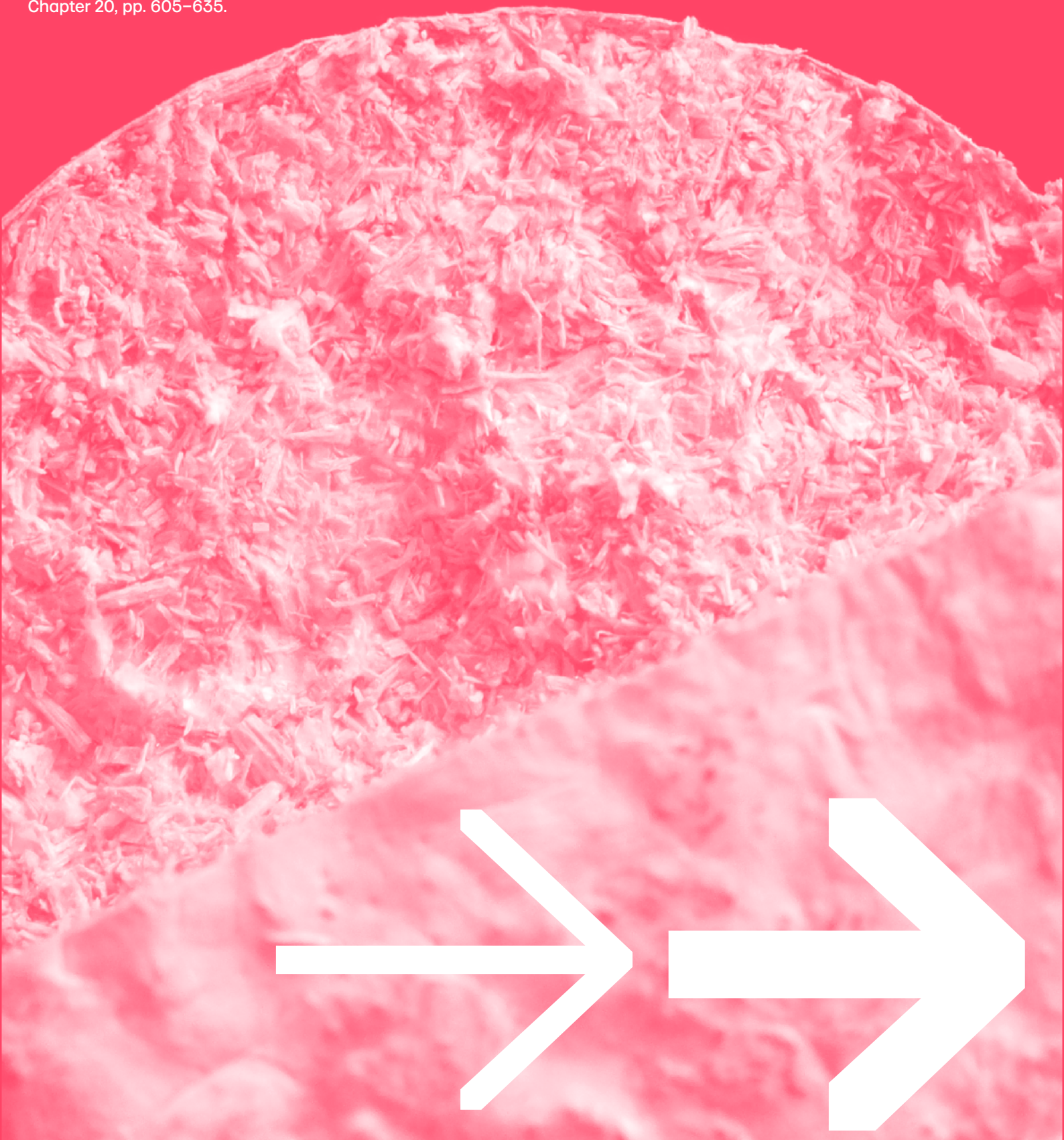
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Design Strategies for Mycelium-Based Composites

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Adrien Rigobello and Phil Ayres

Abstract

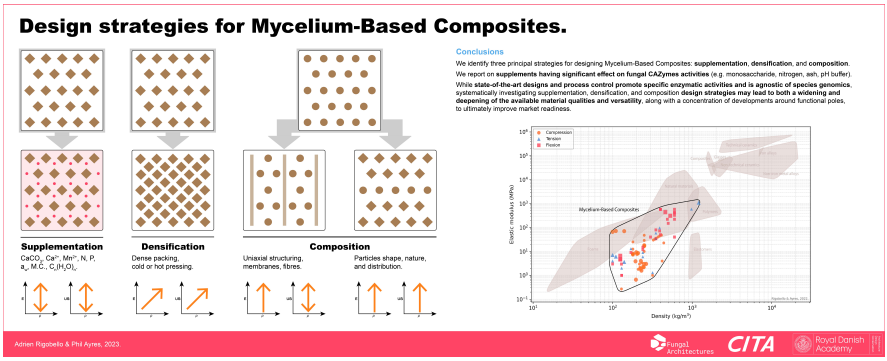
Mycelium-based composites (MBC) can be designed utilising a wide range of lignocellulosic substrates and widely distributed versatile ligninolytic fungi. While a wide range of mechanical behaviour has already been reported in the past 15 years, showing potential to obtain viable products for a variety of uses; no systematic description of the engineering parameters has been established till date. We review carbohydrate-active enzyme (CAZyme) activities of fungal species, lignocellulosic substrate chemical profile at cellular level, wetting characteristics, substrate aggregate and composition characteristics. We identify three principal strategies for designing MBC: supplementation, densification and composition, and discuss them regarding outstanding reports from the state-of-the-art. We report on solid-state fermentation supplements having significant effect on fungal CAZymes activities (e.g. monosaccharide, nitrogen, ash, pH of buffer). State-of-the-art designs and process control promote specific enzymatic activities independent of species genomics; systematically investigating supplementation, densification and composition design strategies in the future may lead to both a widening and deepening of the available material qualities, along with a focus on developments around functional poles. Additionally, future reproducibility studies of MBC development reports may both improve the overall market readiness and public adoption of MBC solutions and valorise the wealth of material and design semiotic properties that the versatility and affordability of MBC systems support.

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Graphical Abstract



Keywords

Solid-state fermentation · Ligninolytic fungi · Mycelium-based composites · CAZymes · Biocomposites · Biomaterials

Highlights

- We identify three principal strategies for designing mycelium-based composites: supplementation, densification and composition.
- We report on supplements and wetting metrics having significant effect on fungal CAZyme activities and mycelia phenotypic expression.
- We report on the composition strategies having significant effect on mechanical response of mycelium-based composites.
- We present state-of-the-art mechanical reports comparatively from species and substrate specifics.
- Systematic investigations in future may lead to widening and deepening available material qualities along with a focus on functional poles.

20.1 Introduction

Inherited from the nineteenth- to early twentieth-century method of fungal strain transfer by lignocellulosic solid-state fermentation (SSF) [1], mycelium-based composite (MBC) cultivation protocols can be based on a wide range of substrates containing lignocellulosic polymers using versatile ligninolytic fungi. Widely available basidiomycota have been instrumental in developing composites with a wide range of mechanical performance, showing potential to obtain viable products for a variety of uses. However, no systematic description of the engineering parameters has been established till date. Although the first MBC commercial venture was established in 2006 (Ecovative Design LLC), an aggressive patent strategy

combined with an inertia in explaining the engineering-driving principles of MBC has resulted in a very low market readiness and market permeability of the material solutions. With a global urgent need to adopt low-energy and low-impact alternatives to existing detrimental material systems on a scale, we argue that rationalising and systematising approaches to analysing MBC systems is necessary to actualise their potential and facilitate market readiness.

As composites, MBC are made of a matrix phase that corresponds to a continuous and foam-like or elastomer-like mycelium, and a dispersed phase composed of particles and/or fibres. Additionally, surface or volumetric components can be used to compose the substrate of a composite and modify its properties. Fungal decay is functionalised as a binding method between elements of the dispersed phase. From this generic definition, we can appreciate the sheer breadth of the material design space, and can identify the main material characteristics for defining the cultivation protocol and specifying composite properties:

- Enzyme arrays of fungal species,
- Substrate chemical profile,
- Substrate wetting characteristics,
- Substrate aggregate characteristics,
- Substrate density and composition.

The resulting composite properties depend on the SSF model that is adopted. While tacit knowledge of MBC craft is sufficient for reaching a satisfactory output, thanks to the competitive colonisation dynamics of the fungi in use, the explanation and modelling of the driving parameters of MBC systems can foster a deeper engagement through engineering practice, thus extending the reach and effectiveness of material research and improving market readiness. Paste-like substrate design for fungal colonisation is not studied at present. The inner structure of a typical MBC is presented in Fig. 20.1.

We start by presenting a review of the MBC model from a ligninolytic fungi enzymatic activities perspective, and proceed to describe the biochemistry of



Fig. 20.1 Typical inner structure of an MBC

lignocellulosic substrates while correlating it to fungal enzymatic abilities. Building upon studies of fungal niches in the wild and their parasitic or saprotrophic behaviour in trees, we come to understand the hydrolytic abilities of wood in relation to their chemical composition and evolutionary stage as a primary parameter for predicting fungal colonisation. A comprehensive review of supplementation methods for influencing enzymatic activities is presented for ultimately adapting protocols to locally available resources. Furthermore, in describing the behaviour of MBC from this perspective, we review studies that have shown a clear correlation between the mycelial mechanical properties and their cultivation substrate chemical composition. While these properties have been demonstrated to be modifiable to result in foam-like or elastomer-like behaviour, nature of substrate aggregates (size, distribution and shape) are the principal contributors of composite strength and stiffness. Outstanding mechanical results from the MBC state-of-the-art are then discussed with regard to these insights, in an effort to describe the behaviour of composites. We identify three main substrate-based exploration strategies for the MBC system: densification (by dense packing, cold- or hot-pressing), composition (by introducing structuring elements) and supplementation (targeting mycelium properties, and based on chemical tuning of the substrate). Furthermore, we have analysed available data in published peer-reviewed studies to offer a critical review of the state-of-the-art, and conclude on the most significant engineering parameters and recommend research directions.

20.2 Enzymatic Activities of Ligninolytic Fungi

The species of *Trametes*, *Ganoderma* and *Pleurotus* are among the most frequently cited fungal families in MBC design [2]. *Schizophyllum commune* is a less investigated species but there is a continued interest [3], especially because it is one of the few mushroom-forming fungi for which genes have been inactivated by homologous recombination [4]. This strategy has proved useful for genetic studies and functional genomics [5]. Gene inactivation articulates the principles of homologous recombination for replacing an existing autologous gene with a designed heterologous gene. Furthermore, the recombinant DNA constructs of *S. commune* has been reported to express in other mushroom-forming fungi, such as *Pycnoporus cinnabarinus*, which supports its importance for fungal biology [4, 6]. *Irpex lacteus* has been used previously in MBC development [7], as in *Agaricus bisporus* [8] and *Fomes fomentarius* [9]. Other less investigated species have also been reported [2].

Wood-rotting fungi can be classified by their decay activity, which cover three main modes: white-rot (specified as selective delignification and simultaneous rot), brown-rot and soft-rot. Across the surveyed species, all of them are white-rot, ligninolytic fungi. They have the enzymatic ability to decompose all three principal polymer classes found in wood: lignin, hemicellulose and cellulose, but most often degrade primarily lignin and hemicellulose as reported in *Ganoderma lucidum* [10]. In such cases, these species qualify as selective delignifies, as opposed to simultaneous decomposers where all three polymer classes endure similar decay

rates. Brown-rot fungi use a two-step oxidative–enzymatic mechanism for the breakdown of cellulose and hemicellulose [11]. Their lignin degradation is limited to modification in lignin, although brown-rot fungi is a polyphyletic group evolved from at least seven white-rot lineages [12]; they reportedly lack more than 60% of the genes known to be present in white-rot species. Brown-rot species degrade cellulose and hemicellulose at a higher rate than white-rot species in laboratory studies [13]. This decay mode is most often encountered in coniferous trees. Soft-rot fungi commonly share the ability to demethylate. This mode is associated with Ascomycota and monokaryotic mycelia, and is distinctive for its tunnelling decay pattern through lignin layers to reach cellulose and hemicellulose-rich areas [14]. Lignicolous fungi can benefit from a variety of enzymatic strategies to ultimately fully decompose lignocellulosic substrates. The environmental conditions for enzyme activation define the species niche and their subsequent distribution. The principal wood-rot enzyme families are presented in Table 20.1.

Lignin linkages result from radical reactions, in contrast to polysaccharide polymerisation resulting from hydrogen bonding. Lignin, therefore, does not involve hydrolytic enzymes and cannot be used as a source of carbon or energy for most wood-rot fungi. Its decomposition serves primarily for the fungus to gain access to hemicellulose, pectin and cellulose compounds. Lignin depolymerisation is mostly initiated by aromatic ring oxidation by peroxidases (by use of H_2O_2 or R–OOH) [35], while other aromatic compounds in the cell structure are mostly catabolised by monooxygenases and dioxygenases. Peroxidases are heme proteins, such as lignin peroxidase (LiP) and manganese peroxidase (MnP). LiPs are known to oxidise phenolic aromatic substrates along with various nonphenolic lignin compounds, and other organic compounds with a high redox potential. MnP leads to H_2O_2 oxidising Mn^{2+} to Mn^{3+}, an oxidising compound to monomeric phenols. Mn^{3+} chelates permeate through wood cells and decay lignin selectively. These chelates cannot oxidise nonphenolic compounds, but radicals formed by Mn^{3+} oxidation can, in turn, oxidise benzyl alcohols and other diarylpropane structures. MnP can also peroxidise lipids [36]. Versatile peroxidase (VP), in addition to MnP and LiP, are enzymes that are Mn^{2+} specific, but that can oxidise phenolic and nonphenolic compounds in its absence (as with LiP). They are involved in oxidation of high redox potential aromatic substrates, thus reducing Mn^{2+}-independent oxidation but not affecting Mn^{2+}-dependent one [15]. Mn^{2+}-independent oxidation capabilities for phenols, small dye compounds and amines has also been described for MnP [35], and it is predicted from sequencing that similar divergent MnP may be present in *Trametes* spp., *Pleurotus* spp. and *A. bisporus* among other species [35, 37]. Hydrogen peroxide, necessary for LiP and MnP activities, is the product of fungal enzymatic activity. A peroxidase family with wide substrate specificity, active at pH of approximately 3, has been identified in fungi: dye-oxidising peroxidase (DyP). In addition to oxidising peroxidase substrates, it can perform on anthraquinones. Laccases are a very versatile family of enzymes found in nearly all fungi including those without lignin degrading abilities [38]. These enzymes perform functions ranging from synthesis of melanin and other pigments to conidia and fruiting body formation, and lignin decomposition [39]. These belong to a multicopper enzyme

Table 20.1 Extracellular enzymes in lignicolous fungi and their optimal activation temperature and medium pH [15–34]

Target polymer	Enzyme family	Reaction	Substrate	Temperature (° C)	pH
Lignin	Lignin peroxidase	Oxidation	Phenol, nonphenolic aromatics	28–40	3.0–4.5
	Manganese peroxidase	Oxidation	Manganese ions, phenol, small dye compounds, amines	23–40	5.0–7.0
	Versatile peroxidase	Oxidation	Manganese ions, phenol, nonphenolic aromatics	35	7.0
	Dye-oxidising peroxidase	Oxidation	High redox-potential dyes, nonphenolic lignin model compounds	28–70	3.0–7.0
	Laccase	Oxidation	Phenol, diphenol, aminophenol, aromatic compounds	35–50	4.0–5.0
Hemicellulose	Xyloglucanase	Hydrolysis	β-1,3-Glucans, carboxymethyl cellulose, xyloglucan oligosaccharides	37–75	3.0–9.4
	Xylanase	Hydrolysis	Glycosic bonds in xylan backbone	28–50	3.0–9.0
	Mannan-degrading	Hydrolysis	Mannan, galactomannan	45–92	2.4–7.5
	Polygalacturonidase	Hydrolysis	Pectin	–	–
	Pectin lyase	β-Elimination	Heavily methyl-esterified	50	8.0
	Pectate lyase	β-Elimination	Lower esterification and Ca^{2+}	30–90	5.0–11.5
	Cellobiose dehydrogenase	Oxidation	Cellobiose, cellodextrins, lactose	60	3.0–10.0
Cellulose	Cellobiohydrolase I	Hydrolysis	Crystalline cellulose	40–80	–
	Cellobiohydrolase II	Hydrolysis	Amorphous cellulose, β-xylan	50–63	4.0–6.0
	GH5 endoglucanase	Hydrolysis	Cellulose	45–65	4.5–7.0
	GH7 endoglucanase	Hydrolysis	Crystalline cellulose	45–65	4.5–7.0
	GH12 endoglucanase	Hydrolysis	Amorphous cellulose	45–65	4.5–7.0
	GH45 endoglucanase	Hydrolysis	Crystalline cellulose	45–65	7.0
	GH1 β-glucosidase	Hydrolysis	β-Linked oligosaccharides, aglycone-linked β-glucosides	25–50	5.0–7.0
	GH3 β-glucosidase	Hydrolysis	β-Glucan	25–50	5.0–7.0

family and catalyse the removal of an electron from phenolic hydroxylic groups to form phenoxy radicals, which then polymerise via radical coupling. Ultimately, the reactions lead to the presence of oxidised quinones and couples oligomeric products. Laccases also have wide substrate specificity and can act upon an extended range of phenols, diphenols, aminophenols and aromatic compounds. When their catalytic reaction is accompanied by demethylation, it may lead to ring cleavage. Tying lignin decomposition to cellulose and hemicellulose decay, cellobiose dehydrogenase (CDH) has the ability to oxidise cellobiose, cellodextrins and lactose [16]. Hemicellulose-degrading enzymes are capable of hydrolysing various polymers, β-1,4-galactosidase degrades xyloglucan, xylan, galactomannan and pectin [17]. More specific xyloglucanases, xylanases, mannan-degrading enzymes and pectinases can be found in all fungi to a varied extent. Typically, pectin can be hydrolysed by polygalacturonidases, but also cleaved by β-elimination by pectin and pectate lyases. Cellulose hydrolysis by fungal enzymes employs glycohydrolases (GH) by glycosic bond cleavage. Among GH families, cellobiohydrolase I (CBH I) is the major cellulase being produced in white- and soft-rots; it binds to crystalline cellulose and hydrolyses it into cellobiose. Cellobiohydrolase II (CBH II), found in all but brown-rot fungi, binds to amorphous cellulose and generates cellobiose from the non-reducing end of cellulose using acid catalysis of the β-1,4-glycosic bond. Endoglucanases are a major enzyme group capable of hydrolysing cellulose polymer from within the molecule rather than its end. The variety encountered in fungi refers to four GH families: GH5, GH7, GH12 and GH45. GH5 endoglucanase is a less studied family in fungi. GH7 endoglucanase displays a similar mechanism to CBH I, with the difference that it can hydrolyse from the middle of cellulose molecules. GH12 endoglucanases lack a carbohydrate-binding module (CBM). They are therefore unable to degrade crystalline cellulose but can hydrolyse it in its amorphous form. GH45 endoglucanase benefits from a CBM. β-Glucosidases, produced by all fungi, belong to GH1 and GH3 families. GH1 β-glucosidases are able to cleave soluble β-linked oligosaccharides from up to nine glucose residue chains and aglycone-linked β-glucosides, and can be competitively inhibited by the products of CDH activity. GH3 β-glucosidases can remove single glucosyl residues from the non-reducing ends of oligo- and polysaccharides such as β-D-glucans, β-1,3-D-glucans, β-1,4-D-glucans [40]. Lytic polysaccharide monooxygenases (LPMO) operating by oxidative cleavage have been shown to enhance cellulose degradation [41, 42].

Carbohydrate-active enzymes (CAZymes) have been classified into six principal families: auxiliary activities (AA), glycoside hydrolase (GH), glycosyltransferase (GT), polysaccharide lyase (PL), carbohydrate esterase (CE) and carbohydrate-binding module (CBM). We have summarised the genome distribution of *Trametes versicolor*, *Pleurotus ostreatus* PC15 and *G. lucidum* G0119 according to this nomenclature in Table 20.2 to exemplify specific species decaying modes. *G. lucidum* benefits from the widest known CAZymes array with 565 genes having been assigned to these functions [10, 44]. This is a well-known versatile fungus capable of selective delignification, as we have previously confirmed experimentally [45]. The actual decay strategies in selective delignification-capable fungi has been

Table 20.2 Genome distribution of CAZymes from selected species: *T. versicolor* (TV) [43], *P. ostreatus* PC15 (PO) [43] and *G. lucidum* G0119 (GL) [44]

CAZyme family	Enzymes	TV	PO	GL
AA	Laccase, LiP, MnP, VP, DyP, CDH, LPMO	89	114	96
CBM	GH5-GH7-GH45 endoglucanase, CBH I	47	82	33
GH	GH12 endoglucanase, β-glucosidase, xyloglucanase, xylanase, polygalacturonidase, mannan-degrading, CBH II	222	235	262
GT	Glycosyltransferases	85	67	72
PL	Pectin lyase, pectate lyase	9	23	11
CE	Carbohydrate esterases	19	28	91

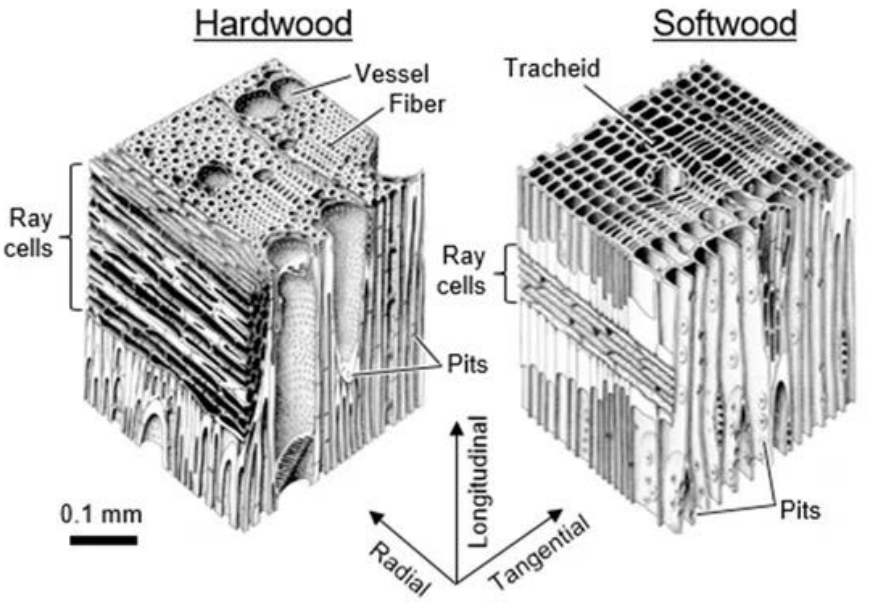
correlated to wood species, lignin contents, temperature, pH and moisture content [46–50].

20.3 Lignocellulosic Substrates

Lignin is the most effective barrier to fungal decay as it prevents migration of low molecular weight diffusible agents (i.e. H_2O_2, veratryl alcohol, oxalate, Fe^{2+}, Fe^{3+}, Mn^{2+}, Mn^{3+}) for decomposing cellulose and hemicellulose. The nature of lignins is of considerable influence on the effectiveness of enzymatic activity. Lignins are polymers of phenylpropene units and they comprise guaiacyl, syringyl and *p*-hydroxyphenyl units. Gymnospermous woods (conifers) are almost exclusively composed of guaiacyl monomers, while angiospermous trees (hardwood) contain approximately equal shares of guaiacyl and syringyl monomers [51]. Similarly, different distributions of hemicelluloses are found in gymnospermous trees (galactoglucomannan, glucomannan and arabinoglucuronxylan are the principal ones) and in angiospermous trees (arabinogalactan, xyloglucan and various glucans are the principal compounds) [14]. Pectins are primarily made of β-1,4-D-galacturonase acid units and their methylesters, interrupted locally by 1,2-linked L-rhamnose units [52]. The primary constituent of wood is cellulose, which is made of repeating glucose units joined by β-1,4-glycosyl linkages. Cellulose represents about 50% of the wood cell wall, lignin 25%, hemicellulose 20–25% and pectin 1–4%. Hemp is often used in MBC practice. Their lignocellulosic components ratio is C:HC:L, which is close to 15:2:1, while their shives display a ratio closer to 2:1:1. The cultivation period and weather has been reported to cause differences in non-cellulosic composition [53].

The variety of wood structures and their chemical composition pairs with various decay strategies driven from the set of fungal CAZyme expression. These wood specifics can be understood from the evolutionary history of cell specialisation. Principal wood structures are presented in Fig. 20.2, reprinted [54]. Wood cells can be made of various specialised cells: tracheids, xylem ray parenchyma, axial parenchyma and vessels. Tracheids are found down to a first evolutionary stage of trees in gymnospermous woods, where they combine the tasks of water transport and

Fig. 20.2 Principal structures of soft- and hardwood; reprinted [54]



structural stiffness. In contrast, gymnospermous woods typically lack vessels and rarely have axial parenchyma. Their tracheid-to-xylem ray parenchyma ratio is most often in the vicinity of 9:1. They contain principally guaiacyl lignin, a product of coniferyl alcohol [55]. In more recent angiospermous woods, water is transported by vessels while tracheids perform structurally. Another evolutionary stage of water transport system in trees can be exemplified by European beech (*Fagus sylvatica*) and birch (*Betula pendula*). Their xylem includes vessels along with tracheids, the former being the only distinct location of guaiacyl lignin. Tracheids and parenchyma contain a fraction of syringyl lignin spread across the cell wall layers, making this secondary evolutionary wood structure attractive for white-rot fungi [14]. A third evolutionary stage can be illustrated by European oak (*Quercus robur*), for which the xylem is ring-porous, containing early and late wood vessels in bands within a thin-walled tracheid matrix. The strength of the wood comes from libriform fibres containing higher syringyl lignin, while tracheids have a higher guaiacyl content [56]. A late evolutionary stage is exemplified by sycamore (*Acer pseudoplanatus*) and ash (*Fraxinus* spp.), where syringyl dominates [57]. Here, vessels are dedicated to water transport, and are jacketed in living libriform fibres that are included amongst dead ones. Guaiacyl lignin has been reported to have a higher resistance to fungal decay [14], which has been corroborated by phylogenetic and experimental studies, showing that catalytic tryptophan presence in more recently evolved peroxidases (located in LiP and VP families) is more efficient at oxidising angiospermous woods, while others perform better in gymnospermous woods [58]. Early-

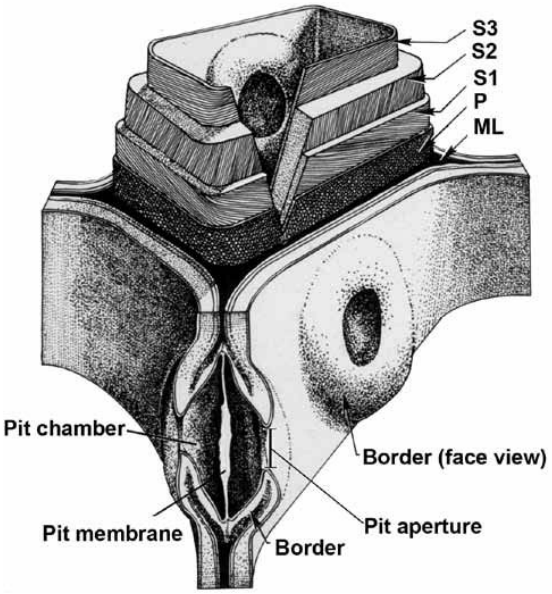


Fig. 20.3 Wood cell wall structure: middle lamella (ML), primary wall (P), secondary wall (S1, S2, S3); reprinted [59]

stage parenchyma decomposition, and a resistance from libriform fibres to white-rot decay, has been confirmed experimentally [14].

The structure of the wood cell wall composing the xylem divides between the middle lamella (ML), primary wall (P) and three layers of secondary wall (S1, S2, S3). They are represented with their fibre orientation and indicative thicknesses in Fig. 20.3, reprinted [59]. The structuring middle lamella is composed of lignin, calcium and pectic compounds that are amorphous. Crystalline cellulose is most prominent in younger wood, so it dominates the S3, and is found in lowering percentages moving towards the P. Inversely, hemicellulose can be found to a greater extent in the P and in reducing shares towards the S3, while lignin is quite evenly spread across the cell wall, with higher concentrations in the ML. The distribution of lignocellulosic compounds in woods is presented in Table 20.3. Colonisation happens through pit apertures present in wood, or the inducing of boreholes by enzymatic activity. Although pectin contents are marginal in wood, pectinases expression is critical to decomposing the pit membrane (Fig. 20.3), which also contains cellulose and allows the fungus to proceed in colonising [61]. Calcium is associated with pectin in wood; Ca^{2+} can combine with fungal oxalic acid and result in pectic acid which solubilises pectin [52]. Calcium, potassium, phosphorus and magnesium are the major elements associated with plant fibre. The mineral content (ash) in softwood is 0.02–1 wt% and 0–5 wt% in hardwood [62]. We can note that large concentrations of calcium oxalate and residues of MnO_2 accumulate during white-rot decay [50]. Moreover, wood can contain a variety of toxic

Table 20.3 Approximated lignin distribution in the different layers of the wood cell wall; adapted [60]

Wood type	Cell type	Layer	Share of total lignin content (%)	Lignin concentration (%)
Gymnospermous early wood	Tracheid	S1–S3	65	24
		ML, P	21	49
		Cell corners	14	64
Gymnospermous late wood	Tracheid	S1–S3	75	22
		ML, P	14	51
		Cell corners	11	78
Angiospermous wood	Fibre	S1–S2	60	19
	Fibre	ML, P	9	40
	Fibre	Cell corners	9	85
	Vessel	S1–S3	9	25
	Vessel	ML, P	2	40
	Ray cells	S1–S3	11	25

compounds, mainly polyphenols or tannins in angiospermous trees and phenolic compounds in gymnosperms. Selective delignification-capable fungi can degrade polyphenols. *Ganoderma* spp. are able to degrade polyphenolic compounds and show chemotropic growth towards these [63]. Nitrogen availability is critical to fungi for enzyme synthesis; the limited amounts in wood makes it resilient to decay (the C:N ratio of birch wood is typically of 55 and 401 in sycamore wood). This compound is mostly found in parenchyma and is known to migrate towards the bark surface during log desiccation [64]. Moreover, wood cell walls specialise depending on local plant mechanical stress and taxonomic order (Fig. 20.4, reprinted [65]), which may lead to a more specific consideration of lignocellulosic substrates to establish cultivation protocols.

The wide distribution of wood-decaying fungi suggests, with increasing evidence, that local environmental factors and substrate specifics drive and regulate colony extent between the various fungal species present. Furthermore, the pH of wood is most often in the 4.0–5.5 range (beech: 5.11, birch: 5.29, for instance) [66]. It has been demonstrated that substrate pH is also influenced by the action of decay [46]. It has been reported, by comparative transcriptonics, that white-rot related genes of a *Phanerochaete carnosa* species would all express in fir (*Abies balsamea*), pine (*Pinus contorta*), spruce (*Picea glauca*) and maple (*Acer saccharum*), but to a varied extent. For instance, MnP genes read was 2.42 times higher in fir compared to maple, but LiP-related genes read was 4.7-fold higher in

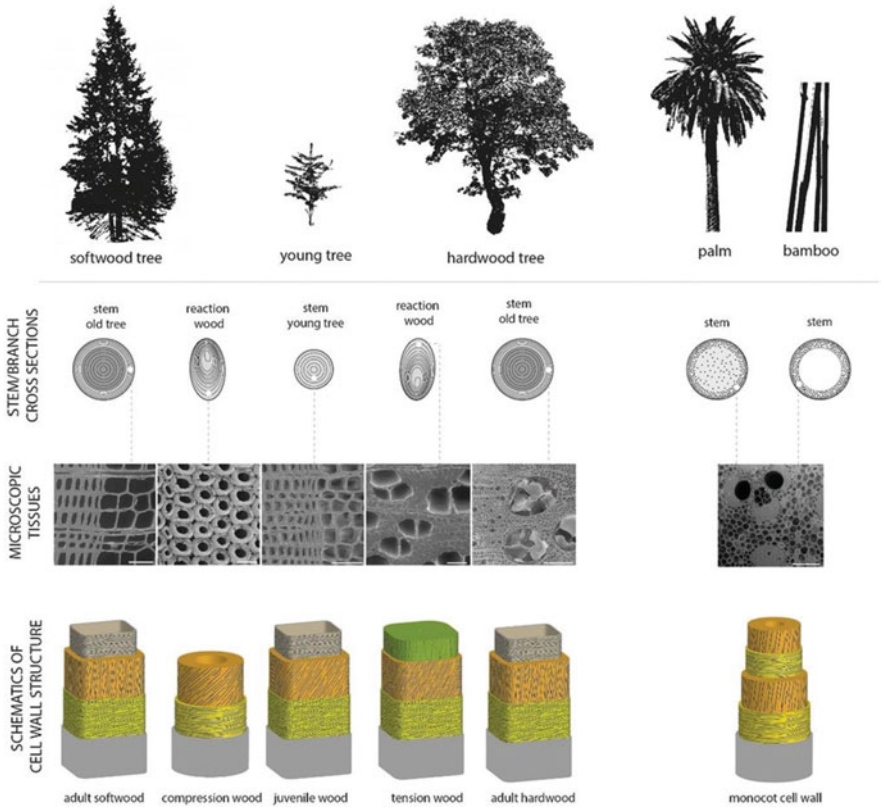


Fig. 20.4 Structure of arboreal plants; reprinted [65]. Microscopic tissues from left to right: adult spruce wood (bar: 50 μm), spruce compression wood (bar: 20 μm), juvenile spruce wood (bar: 50 μm), poplar tension wood (bar: 50 μm), adult oak wood (bar: 200 μm) and vascular bundle of bamboo (bar: 200 μm). Last row showing schematics of cell wall structures. Light grey indicates the compound middle lamella (0.5–1.5 μm) and primary cell wall (approximately 0.1 μm); yellow: S1 layer with a thickness of 0.1–0.35 μm; orange: S2 layer as 1–10 μm thick; inner light grey layer shows the 0.5–1.1 μm thin S3 layer; the green G-layer in tension wood fibres can fill the whole lumen and the cellulose fibrils are oriented along the longitudinal cell axis

maple [67]. This indicates both the resilience of species to the use of a variety of vernacular substrates, but also a precariousness considering the sheer distribution of competitors of an individual species. This can lead to fungi chemically wrestling for resource [68]. *Ganoderma* spp. have been recognised as having the competitive advantage of being able to degrade reaction zones in trees [63]. Studies of lignicolous fungal ecological distribution point to water content and temperature variations being the principal drivers of selection [69]. Climate change has then been related to shifts in fungal niches across hosts [70]. In the wild, host tree species and their evolutionary stage can have an influence on patterns of decay; the tree group that has the most diffuse xylem vessel system may result in a slower initial

colonisation. Similarly, gymnospermous wood having 3–4 mm long tracheids benefits from a cellular compartmentalisation strategy to reduce fungal decay, despite its guaiacyl lignin content. These topological effects are prominent at the early stage of fungal colonisation [64].

20.4 Water in Fungal Decay

The presence of capillary water in the substrate is a prerequisite to allow extracellular transport of metabolites in fungi, and to establish osmotic potential for allowing turgor pressure and subsequent hyphal growth. Turgor also enables fungi to penetrate solid substrates [71]. It has been reported that the radial growth of white-rot species *Physisporinus vitreus* at pH 5 and 25 °C increased from just below 0.5 mm/day at $a_w = 0.97$ to 4.5 mm/day at $a_w = 0.998$ [72]. The production of enzymes by fungi and their catalytic activity has been related to water activity with varying local optima depending on the depressor that was used in the study, pointing to $a_w \geq 0.99$ for optimal catalytic activity here [73].

SSF requiring the absence of solution in between particles for a mycelium to develop, the substrate water holding capacity is an important parameter for ensuring an optimal fungal decay and growth [11, 73, 74]. In timber, moisture can exist as bound water within cell walls below the fibre saturation point (FSP), free liquid water in cell cavities above FSP and water vapour. Relative humidity (RH) of up to 97–98% is generally considered to correspond to cell wall uptake (hygroscopic range), and as RH moves towards saturation, wood cell void uptake tends to become more significant in the moisture distribution (over-hygroscopic range) [75]. The latter is conventionally considered as an optimal range for optimal fungal growth in lignocellulosic SSF. The validity of this convention is nonetheless discussed as it was shown that *T. versicolor* would colonise European beech wood at a minimum moisture content (MC) of 15% for a mass loss over 2% (equivalent to 75–80% RH at −1–37 °C); [76]. Brown-rot fungi grow at 92–97% RH, and that instrumental limitations are encountered in the over-hygroscopic range, thus complicating measures and experimental control [75]. For instance, the FSP of European beech wood has previously been determined to be 24.02% MC at 97% RH and 20 °C [77]. The minimum moisture threshold has been introduced as the metrics having the most significant impact on predicting fungal growth [11, 78]. It represents the MC below which fungal decay cannot be initiated, and is typically given at an mass loss 2%.

Available hydroxyl groups are the predominant sorption sites for water molecules in wood. Their presence in hemicellulose is twice as high as in lignin and four times higher than that of cellulose [79]. It has been reported that cellulose can endure structural changes while wood is desiccated [80, 81]. In the literature, it was hypothesised that lower desiccation rates would give enough time for amorphous cellulose to crystallise, rendering the substrate less hydrophilic. On the contrary, higher rates of desiccation may lead to more prominent amorphous cellulose and more hydroxyl availability. Lignin, hemicellulose and pectin interrupt cellulose

crystallisation, which reflects on the higher crystalline content towards the S3 cell wall layer [82]. Because hemicellulose has a softening point at room temperature and 18–20% MC [74], while lignin softens above 60 °C at a minimum of 15% MC [83], and cellulose is hardly softened in its crystalline conzuration (200 °C), higher thermal treatments may result in hemicellulose degradation, thus rendering the substrate less hydrophilic [84].

Finally, the amount and quality of water in the SSF system varies during fermentation. Fungal oxidative decay of polysaccharides frees water along with CO_2:

$$(C_6H_{10}O_5)_n + 6n(O_2) = 6n(CO_2) + 5n(H_2O) \quad (20.1)$$

where $C_6H_{10}O_5$ is the repeating unit of glucose-based polymers, such as found in starch or cellulose. As bound water is also released during decay, and while substrate dry mass loss occurs, the MC increases if the system is closed. The activity and location of water in SSF systems are are then necessary information to predict fungal development. They are of primary interest to design and produce MBC. To the best of our knowledge, the role of water in MBC production has not yet been reported. The optimal moisture content for laccase, LiP and MnP activation is consistently reported to be 60–75% [85–88].

20.5 Regulating Enzymatic Activities

MBC protocols can be adjusted based on the chemical profile of locally available wood resources and selected fungal species to achieve various design goals. This an be approached by pH and temperature regulation, and monosaccharide, nitrogen and metal ion supplementation. Optimal ligninase activity was reported at pH 4.0–5.0 and 30–35 °C [85, 86]. The pH regulation can be achieved by the addition of $CaCO_3$ in the substrate as a buffering agent to avoid acidification from CO_2 production. Gypsum supplementation has been reported in MBC cultivation [89]. While its use has no impact on the substrate pH, it contributes to adding calcium to the medium. This is useful for pectate lyase activity if the wood has a low Ca^{2+} content, which acts on the 1–4% share of pectin in wood. Varying the C:N ratio has also been shown to lead to an optimum of 20:1 [85]. The form of the C- and N-rich additives has been reported to have an impact on specific enzyme activation. Maltose and urea addition was reported to increase LiP and MnP production in a *T. versicolor* species, while glucose and yeast resulted in increased laccase production [87]. A similar investigation with a *G. lucidum* reported that the use of beef extract increased LiP production by approximately 30% as compared to using yeast extract, peptone or urea, and sucrose had a significant effect on ligninase activity as per other sugar types [85]. Wheat bran is a frequently used substrate additive in SSF. It contains high contents of β-glucans, xylan, cellulose, lignin, B vitamins and mostly minerals (magnesium, phosphorus, manganese, zinc and iron) [90]. The success of this supplement [91] may be related to its mineral contents, especially manganese as it

is a limiting compound for MnP and VP activity [92]. Supplementation of substrates by $MnSO_4$ has therefore been shown to result in an increase in ligninase activity in *S. commune* [86] – most probably by synergistic action of enzymes. Biofuel production aims at efficient extraction of sugars from wood. Selective delignification capable fungi have been extensively investigated in this context. The addition of phenols has been reported to be considerably promoting ligninase activity [93]. Syringic acid has been shown to increase LiP and laccase activity, and gallic acid for MnP in the same study. With an optimised substrate, the authors reported increase of 2.4 to 45.5-fold across the diverse ligninase activities, leading to a 2.3-flod improvement in lignin decay and 7.1 in selectivity value (selectivity is expressed as the ratio of lignin degradation over cellulose and hemicellulose degradation) [94]. The idea of highly selectively degrading lignocellulosic substrates is a very promising track for MBC research. The addition of H_2O_2 to the substrate, necessary for ligninase activity, has also been investigated with significant results [95, 96]. $CuSO_4$ supplements have been shown to increase laccase activity [93]. The use of surfactants such as Tween-80 has been reported to increase ligninase permeation through the cell wall [97], and enhance the solubilisation of lignin and improve hydrolytic activities [98], and even proved useful more recently in promoting all ligninase activity, at a 4 wt% optimum for *G. lucidum* fermentation [88]. With an approximate 80% content of starch, 12% proteins, 3% of other monosaccharides (arabinose, xylose, galactose, glucose) and traces of ash (minerals) [99], white and whole wheat flour constitute useful additives to promote cellulase and hemicellulase activities. Moreover, the use of spent coffee ground is a widespread decay promoter in MBC craft. It is composed of 39% hemicellulose (predominantly mannose and galactose), 24% lignin, 17% proteins, 2% fat and 1.3% ashes. Although it has a high lignin content, coffee grounds contribute to increasing nitrogen and mineral content in the substrate, additionally to glucose polymers. Coffee silverskin, a by-product of the roasting industry, has a quite different chemical profile: 24% cellulose, 29% lignin, 19% proteins, 17% hemicellulose (primarily xylose) and a remarkable 5% ashes [100]. Phenolic compounds present in both substrates have been investigated for their antifungal abilities against white-rot fungi. Using a *T. versicolor* species, it was reported that the supplementation of 3% coffee silverskin extract resulted in about 65% growth inhibition, while 1% caffeine led to a complete inhibition and 1% spent coffee ground in about 20% inhibition [101]. This is related to the introduction of phenolic antioxidants such as caffeine, caffeoylquinic acid lactone and feruoylquinic acid [102–104]. Furthermore, laccase treatment of lignocellulosic substrates has been shown to increase lignin antioxidant activity [105]. Finally, the addition of phosphorus and calcium in the form of superphosphate ($CaH_6O_8P^{2+}$) has proved to increase decay yield. Varying the supplementation from 0.025 wt% to a 0.3 wt% optimum in wheat straw, lignin degradation increased by a 4.5-fold, while cellulose, hemicellulose and protein degradation increased by a factor of about 2.4, while enduring *P. ostreatus* decay [106]. A higher optimum of 0.5 wt% superphosphate addition has been reported for alkaliphilic white-rot *Coprinus* spp. [107], and even 1.0 wt% with *T. versicolor* [108].

20.6 Mycelium Mechanical Properties

Mycelium acts as a binder between wood particles. It does so by degrading lignocellulosic compounds and producing a dense hyphal network that interlocks dispersed particles. The binding between two particles is not distinctly strong, but the very high redundancy in the composite can result in a mechanically valuable material. It has been reported that a *P. ostreatus* mycelium would be more than two times stiffer than a *G. lucidum* one when both cultivated on a cellulose medium [109]. Authors comment on this difference with regard to the higher protein and lipid content in *G. lucidum* that can act as a plasticiser. Furthermore, in this study a set of experiments grown on a PDB-cellulose medium has been compared to the previously mentioned results, and it was found to decrease the Young's modulus of the *G. lucidum* mycelium film by a factor of 3 while increasing its elongation at break by 2.3-fold. This corroborates their previous comment on plasticisers, and suggests that nutrition medium tuning can help composite behaviour. This study suggests that the matrix phase of the composite can be made closer to an elastomer or stiffened by varying protein, lipid or alcohol contents. Similar findings have been reported [111]. Isolated mycelium mechanical behaviour has been investigated in order to model it [112]. This referenced study relies on the characterisation in tension and compression of a pure mycelium. At the resolution of hyphae, researchers have identified three main stages of response of a ligninolytic fungal mycelium: linear elastic behaviour at small strains (hyphae bending); fibre buckling and local structural collapse or densification at larger strains; rapid stiffening associated with full compaction and large number of inter-fibre contacts. To fit their finite-element analysis simulation model, a hyperelastic model was used in this study as mycelium undergoes large displacements in its elastic behaviour and results in a non-linear stress-strain response. Young's moduli are situated in the 0.6–2.0 MPa range for a density in 30–50 kg/m^3, the yield stress in 40–80 kPa and ultimate tensile strength in 100–300 kPa [110]. The results are presented in Table 20.4.

Table 20.4 Tensile behaviour of mycelia from *P. ostreatus* (PO) [109], *G. lucidum* (GL) [109] and an undisclosed species (ND) [110]

Tensile behaviour	PO[a]	PO[b]	GL[a]	GL[b]	ND[c]
Young's modulus (MPa)	28	17	12	4	0.6–2.0
Ultimate strength (MPa)	0.7	1.1	1.1	0.8	0.1–0.3
Elongation at break (%)	4	9	14	33	–

[a] Cellulose medium.
[b] Cellulose and PDB medium.
[c] Undisclosed medium.

20.7 Substrate Mechanical Properties

Most commercially valuable woods have a Young's modulus of 5.5–15.7 GPa at 12% MC, and 4.4–12.3 GPa when freshly cut (green) [113]. The distribution of these performances is similar in gymnospermous and angiospermous woods. Wood particles are therefore important load-carrying members of the composite system as compared to mycelia elasticity discussed previously, and reduce the magnitude of stress experienced by the mycelial matrix. The angular shapes of wood debris contribute to crack initiation and the dewetting behaviour of the larger particles is a principal contributor to damage nucleation in such a two-phase particulate composite, although they contribute to composite stiffening by matrix shielding. Furthermore, the shape, nature and distribution of particles in two-phase particulate composites has been shown to have a substantial influence over the load transfer between members, and hence their overall stiffness [114]. Moreover, while lignin is a primary contributor of strength parallel to grain, hemicellulose supports compression strength perpendicular to grain. Its decay greatly affects the structural integrity of wood and its hardness [115].

In particle studies of granular materials, shape parameters such as flakiness/flatness (ratio of particle thickness to width), elongation (length-to-width ratio), sphericity (deviation from a sphere geometry) and roundness/angularity (measure of angular sharpness) have been investigated [116]. Without matrix phase (the stiffest reported pure mycelium of a *P. ostreatus* strain in 20.6 had a Young's modulus of 28 MPa, 178–534 times more elastic than woods), studies have reported that a 3:1 ratio of flaky to bulky particle content would be an approximate optimal for shear strength (depending on the system of study). This was related to cohesion increase under stress due to particle interlocking. Furthermore, it was reported that a higher particle angularity leads to a decrease in elastic modulus, while ultimate strength increases. The shear strength was reported to follow the same dynamic, increasing with particle angularity. Increasing flakiness and angularity leads to increased cohesion and subjection to higher abrasion. This leads to damage accumulation under repeated loads, thus leading to strain accumulation [116].

While wood particles are of common use in MBC, other more elastic substrates have been used such as nonwoven cotton fibres [117]. A few studies have investigated the addition of non-organic aggregates to a lignocellulosic substrate to improve its stiffness, such as with carbonate sand [118] (which also acts as a pH buffer being mainly composed of $CaCO_3$) and sand and gravel [9]. It was reported that supplementation of a cotton and wheat bran medium with 37.5 wt% carbonate sand resulted in a 1.6-fold stiffening, and a factor 4 increase in ultimate compressive strength, while density increased by 27%.

20.8 Composite Mechanical Properties

The large number of influential variables on MBC mechanical properties leads to the emergence of a variety of methods for tuning them. While common practice leads to an elastic modulus of an approximate maximum of 2 MPa, post-cultivation heat-pressing has been reported to lead to moduli in the range of 35–97 MPa, and flexural moduli in the range of 34–80 MPa [117]. The use of particle-additive strategies, additionally to a wheat bran supplemented medium made of cotton stalk, has proven to have a significant effect: an elastic modulus of 48.5 MPa was reported with 37.5% carbonate sand addition with *P. ostreatus*, while a non-supplemented control in this study reached 30.3 MPa [118]. A second study reported an elastic modulus situated in the 39–60 MPa range for a density in 240–265 kg/m^3 [7]. Authors of the latter are elusive about production details, but nonetheless indicate the use of Alaska birch shavings (*Betula neoalaskana*), millet grain, wheat bran, calcium sulphate and the addition of a natural fibre. The specifics of the performative batch are not disclosed, but a pre-colonisation was performed prior to moulding. The performance increase does not come from a single factor considering the experimental plan. The considerable use of fibres (of undisclosed quality), grains, brans and $CaSO_4$ leads us to understand this as a hybrid improvement using both substrate-based reinforcement with the fibres and additives. Other strategies have been investigated, such as with the introduction of SBR Latex, a bonding agent for the construction industry. With a 5% addition of SBR Latex and 0.5% silane coupling agent to a cotton seed hull medium, the cultivation of a *P. ostreatus* species displayed a 2.39 factor increase in ultimate strength with a 27.6% density increase [119]. The resulting environmental impact has not been discussed in the study. The addition of another material with high environmental impact [120], cellulose nanofibrils (CNF), has been reported [121]. Between two specimen groups, the optimal supplementation of CNF was reportedly 2.5–5 wt%, with a contribution in Young's modulus of a factor 2.56–5.6 and 2.8–7 in ultimate strength. In this study, the effect of densification on composites embedding a 2.5 wt% CNF part was reported: densifying from 300 to 600 kg/m^3 contributed to increase the yield strength exponentially to a factor 17, while the stiffness increased 14.38-fold.

An extensive study of the properties in tension of densified MBC reported that elastic moduli situated in a range of 3.0–13.0 MPa; these values resulting in 24 days cultivated *T. versicolor* mycelia on wheat bran supplemented beech sawdust [117]. For this specimen group, a flexural modulus of 9.0 MPa was reported. The state-of-the-art in MBC research largely considers monolithic and homogeneous composites, besides a few study groups investigating jute type materials in sandwich composite reinforcement and composition with wood panels [45, 122–124].

Across the state-of-the-art, we can identify three main research strategies: *densification* (by composite packing, cold- or hot-pressing), *composition* (by introducing structuring elements such as fibres and aggregates) and *supplementation* (by chemical tuning of the substrate). These are substrate-focused strategies, targeting mycelium properties tuning in the case of supplementation, while the very choice of the fungal species can also result in radically different cultivation lead time and

mechanical properties. Genetic modification of fungi could promote specific enzymatic activity or modify the composition of its cell wall to help improve performance. However, the investment cost and ethics of this practice may lead to prioritising more energy-efficient and affordable techniques.

20.9 Analysis of the State-of-the-Art

So as to analyse the effect of strategies in the MBC state-of-the-art and to present a comprehensive mechanical review for it, we collected reports of elastic moduli and ultimate strength in compression, tension and flexion, along with principal substrate information, species and density [7, 45, 117, 118, 121, 122, 124–135]. There are 105 data points gathered from 18 journal and conference articles. These include articles reporting on all or some of these metrics: strength and/or stiffness in compression, tension and/or flexion; 19 data points had no density reported. In the following data visualisations (Figs. 20.5, 20.6, 20.7, 20.8, and 20.9), only the reports with sufficient data are rendered.

The distribution of species to substrates reports is heterogeneous. For instance, we can observe in Fig. 20.5 that no peer-reviewed report has been made of hemp cultivated *G. lucidum* as per mechanical behaviour. This figure does not consider all substrates and supplements included in MBC, only the primary one.

We present the specific moduli and strengths as per substrates in Fig. 20.6, and specific moduli and strengths as per species in Fig. 20.7. Larger marker size indicates

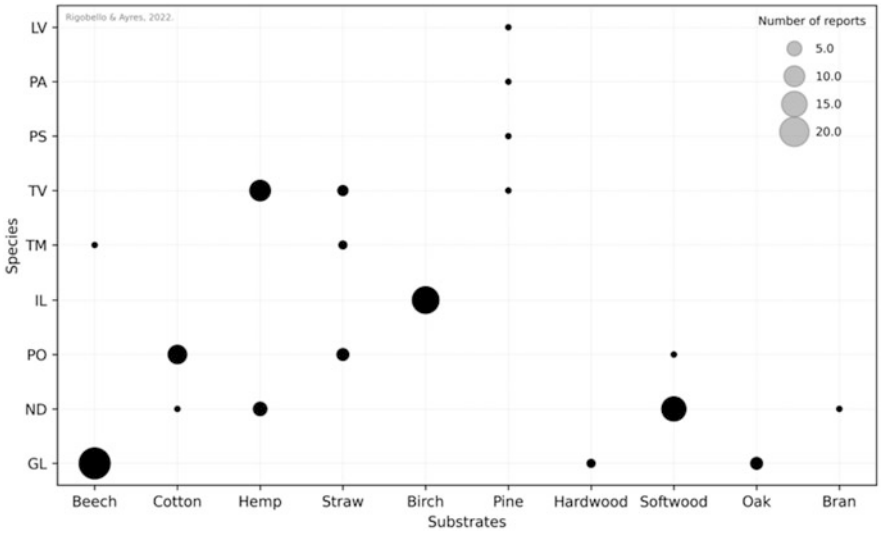


Fig. 20.5 Reports of principal substrates and species. GL: *Ganoderma lucidum*, ND: undisclosed, PO: *Pleurotus ostreatus*, IL: *Irpex lacteus*, TM: *Trametes multicolor*, TV: *Trametes versicolor*, PS: *Pycnoporus sanguineus*, PA: *Pleurotus albidus*, LV: *Lentinus velutinus*

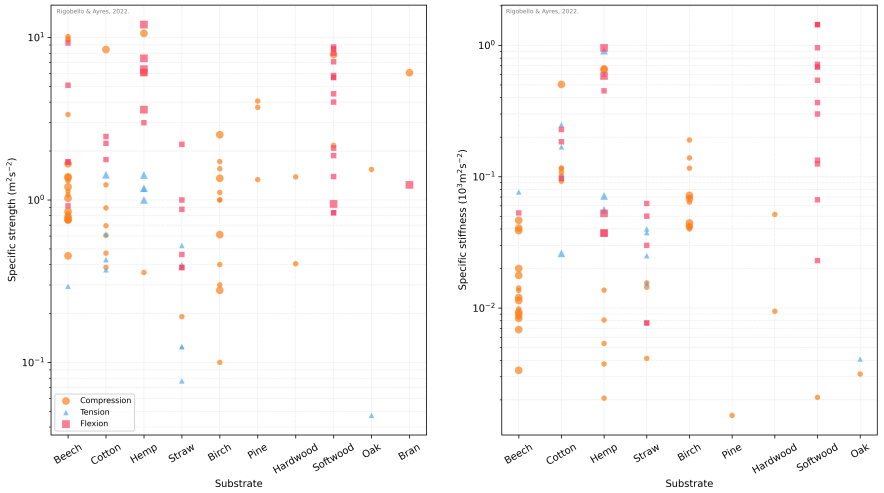


Fig. 20.6 Specific strength and stiffness per substrate. Larger markers correspond to composition strategies

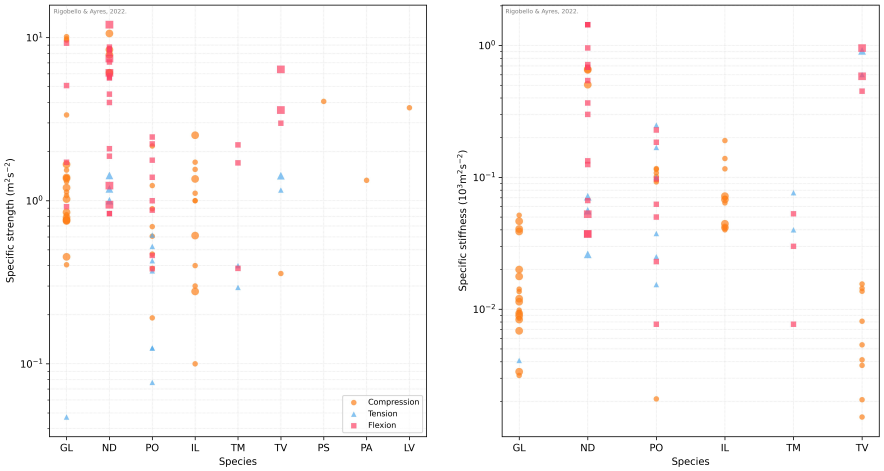


Fig. 20.7 Specific strength and stiffness per species. Larger markers correspond to composition strategies

a composition strategy. We classified MBC having structuring fibres, textiles, inorganic particles, CNF and long fibre balls. Only a few reports on MBC tension modulus (14 data points) and strength (15 data points) have been published. From biochemical review presented above, we can be surprised by the mechanical

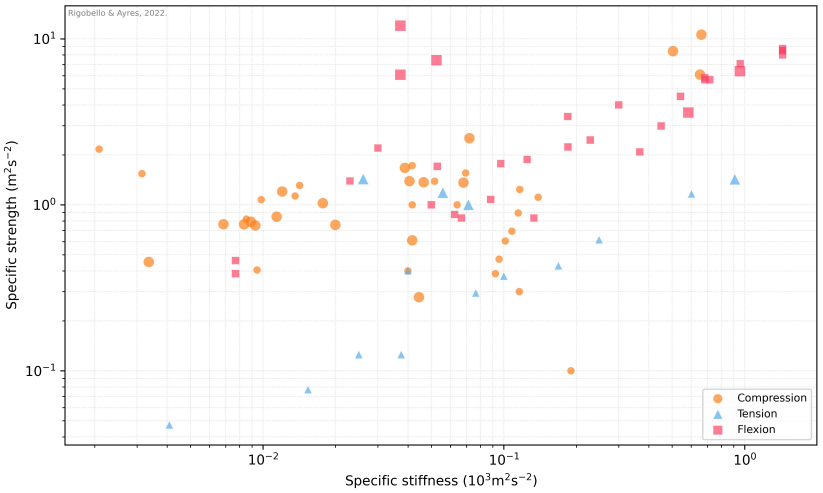


Fig. 20.8 Specific strength as a function of specific stiffness. Larger markers correspond to composition strategies

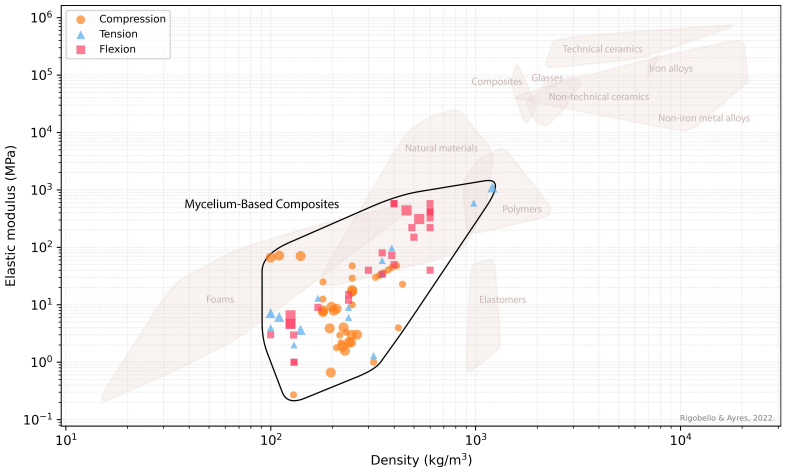


Fig. 20.9 Elastic modulus as a function of density. Larger markers correspond to composition strategies

competitiveness of softwood in Fig. 20.6. It should be noted that these specific reports are of CNF-embedding MBC, thus supplementing softwood with cellulose that is accessible to the fungus without a lignin barrier, having been pressed at an undisclosed pressure at 180 °C for 15 min [121]. Densification of lignocellulosic particles by hot pressing may result in an increased mechanical interlocking of the particles, along with chemical binding from solubilised compounds [83]. Furthermore, in Fig. 20.7, we can identify the undisclosed species in use to be the most

performative with regards to specific stiffness and specific strength. In Fig. 20.8, we present the specific strength as a function of specific stiffness in compression, tension and flexion. We can notice in this figure a departure from the state-of-the-art of a specific study in compression [124]. The specimens of the latter were embedding a surface-binding fabric and packed cotton-ginning waste and hemp pith of undisclosed qualities. The reported specimen densities are of 100–140 kg/m^3. Without further details on the cultivation method, and with the tests having been conducted with only 3–4 replicates, we can hardly conclude on the causes having led to these outstanding values. With critical methodological details being undisclosed, this study is irreproducible and, as such, serves solely as advertisement material.

We present an Ashby map of state-of-the-art MBC data for elastic modulus as a function of density in Fig. 20.9. We can notice that MBC represent a large functional area, mostly situating in between foam and elastomer materials, with reports extending the group span to performances typical of natural materials – the upper-right cluster of flexural modulus reports values typically stiffer than cork and leather. Punctual reports of tensile modulus in the upper-right MBC area would be of a similar stiffness than polyethylene (PE) or PE polymers. The span of the MBC area reflects the three development strategies: densification leads to clear tendency towards increasing the stiffness and strength of MBC for construction materials; supplementation is less distinct at the moment in the reports, but the three data points in the upper-left MBC area are representative of its effect as shown in Fig. 20.8; composition strategies are distributed across the MBC area and reflect a wide variety of inserts and resulting mechanical behaviour. Composition has been demonstrated to lead to significant effect on MBC behaviour in failure control or stiffening for instance [45].

20.10 Conclusions

This review focuses on three aspects: fungal identity (fungal species enzyme array), nutrients (substrate chemical profile and wetting characteristics) and substrate architecture (substrate aggregate characteristics, density and composition). From the enzyme review, we have come to understand that the substrate fitness and mechanical behaviour of mycelia can be adjusted by supplementation. This can be achieved by pH and temperature regulation, and monosaccharide, nitrogen, metal ion and $CaCO_3$ supplements could be systematically investigated in the future as per initial substrate pH and its variation during fermentation. Lignocellulosic substrate selection can also have a significant effect, with non-supplemented softwoods being chemically unfit for optimal white-rot decay. The use of sycamore or ash wood could be investigated in the future as they display higher syringyl monomer contents. The C:N content of sycamore being 401, supplementation may be necessary for optimal fermentation. Furthermore, incubation temperature is commonly fixed throughout the fermentation process below 30 °C, which promotes specific enzymatic activities (LiP, MnP, DyP, xylanase, pectate lyase, GH1 β-glucosidase and

GH3 β-glucosidase). While investigating exothermic heat generation and dynamic temperature control related to species, genome distribution may lead to both a widening and deepening of the available material qualities and versatility.

The effect of water activity upon MBC colonisation-related performances, such as mechanical response, has not yet been reported in the state-of-the-art. Considering reports in SSF studies of several fold increase in colonisation rates, investigating the effect of water activity in MBC may result in longer production times and performance. Moreover, qualifying the effect of hydroxyl group availability in hemicellulose and cellulose, depending on wood processing history, could lead to further medium optimisation. We expect this particular aspect to be specifically useful for paste-like substrate design, such as that found in mycelium-related 3D printing.

Of course, stimulating the focus of future research towards such specific aspects may result in a higher entry barrier for a technique that has a strong interest for its affordability. The use of Fourier transform IR or other spectrometric process by experts within a network of amateur practitioners could be useful to adapt cultivation strategies constructively to local contexts; the qualification of an available substrate may lead to specifying necessary processing, supplementation, species selection and/or relevant composition strategies. While industrial and academic research may result in defining a panel of protocols for the selection of prospective materials, situated citizen craft practices based on locally available materials could coexist.

Across the MBC state-of-the-art, we have identified three principal substrate-oriented design strategies: densification, supplementation and composition. The latter covers a variety of strategies, from CNF introduction to uniaxial reinforcement. These will need to be further investigated to allow an in-depth understanding and serving a goal-oriented material development. It is important to note that most experimental methods are heterogeneous, may not be statistically significant or only partially disclosed. A structured approach to developing material designs along with assessing them will most likely result in the concentration of the state-of-the-art around functional poles in the future, and will help to drive material development in depth. Additionally, future reproducibility studies and critical methodological reviews of MBC studies may both improve the overall market readiness and public adoption of MBC solutions. This can further valorise the wealth of material, design and semiotic properties that the versatility and affordability of MBC systems support.

In this chapter, we have identified the principles driving MBC design and engineering, and have attempted to make these comprehensible. Beyond this focus on the design and production, MBC engage broader discourses related to resource sourcing, circular economy and sustainability. Because MBC have a higher vernacular potential [136], they can be made of recycled substrate, or materials locally and seasonally available. MBC also benefit from unique aesthetics as a result of the phylogenic expression of fungi through craft and substrate design [137]. We hope that the on-going uncovering of the principles for predicting MBC performance contributes to an engineering practice that may satisfy industrial needs (which detrimentally often actualises in patents [138]), but also fosters a rich, open, sustainable and creative citizen craft.

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